Chapter 3. GROUPS

3.3 Cyclic Groups

Today we will investigate

the cyclic subgroup generated by an element.

The cyclic subgroup generated by an element

Theorem 3.25. *Let g be an element of a group G.*

(i) **If the cyclic subgroup $\langle g \rangle$ is finite**, *then there exists a smallest positive integer n such that $g^n = 1$. In such a case:*

(a) $g^k = 1$ *if and only if* $n|k$.

(b) $g^k = g^m$ *if and only if* $k \equiv m \pmod n$.

(c) $\langle g \rangle = \{1, g, g^2, \ldots, g^{n-1}\}$ *and* $1, g, g^2, \ldots, g^{n-1}$ *are distinct elements.*

(ii) **If the cyclic subgroup generated $\langle g \rangle$ is infinite**, *then:*

(d) $g^k = 1$ *if and only if* $k = 0$.

(e) $g^k = g^m$ *if and only if* $k = m$.

(f) $\langle g \rangle = \{\ldots, g^{-2}, g^{-1}, 1, g, g^2, \ldots\}$ *and all these powers of g are distinct.*

Proof.

(i) Suppose that $\langle g \rangle$ is finite. Then the powers $g, g^2, g^3, g^4, \ldots$ can not be all distinct, so we can find two positive integers k and m such that $g^k = g^m$. Assume, for example, that $k < m$. Then $g^{m-k} = 1$, where $m-k > 0$. Then by the Well-Ordering Axiom there exists a smallest positive integer n such that $g^n = 1$.

(a) $g^k = 1 \overset{?}{\Longleftrightarrow} n \mid k$.

Assume first that $n \mid k$. Then $k = qn$ for some integer q.

$$\underline{g^k} = g^{qn} = (g^n)^q \overset{g^n=1}{=} 1^q = 1 /$$

Conversely, if $\underline{g^k = 1}$. Applying the Division Algorithm we have that

$$\underline{k = qn} + r, \text{ with } 0 \le \underline{r < n}.$$

But then $r = k - qn$.

$$g^r = g^{k-qn} = g^k (g^n)^{-q} \underset{g^k=1}{\overset{g^n=1}{=}} 1^k \cdot 1^{-q} = 1 /$$

We have proved that $g^r = 1$, where $r < n$ and n was the smallest positive integer with this property, so $\underline{r = 0}$. Thus:

$$K = qn \Rightarrow \underline{n \mid K}.$$

(b)

$$g^K = g^m \overset{?}{\Leftrightarrow} K \equiv m \pmod{n}$$

$$g^K = g^m \Leftrightarrow g^{K-M} = 1 \overset{(a)}{\Leftrightarrow} n \mid (K-m) \Leftrightarrow K \equiv m \pmod{n}$$

(c)

$$\langle g \rangle \overset{?}{=} \{1, g, g^2, \dots, g^{n-1}\}.$$

Clearly, $\{1, g, g^2, \dots, g^{n-1}\} \subseteq \langle g \rangle$. To show the other containment we need to check that $g^K \in \langle g \rangle$. To do so, we apply the Division Algorithm to write $K = nq + r$, with $0 \leq r < n$.

$$g^k = g^{nq+r} = (g^n)^q g^r \underset{g^n}{=} g^r \underset{0 \le r < n}{\in} \{1, g, \dots, g^{n-1}\}$$

If $g^k = g^m$ for some $0 \le k \le m < n$. Then

$g^{m-k} = 1$, and since $0 \le m-k < n$, and n was the smallest integer with this property, then $\underline{m-k=0}$, that is, $\underline{\underline{m=k}}$.

(ii) Assume now that $\langle g \rangle$ is infinite.

(d) $g^k = 1 \overset{?}{\iff} k = 0$

Clearly $g^0 = 1$. If $g^k = 1$ with $k \neq 0$, then $g^{-k} = (g^k)^{-1} = 1$ and this would imply that $\langle g \rangle$ is finite, a contradiction.

(e) $g^k = g^m \overset{?}{\iff} k = m$

$$g^k = g^m \iff g^{k-m} = 1 \overset{(d)}{\iff} k - m = 0 \iff k = m$$

(f) By definition

$$\langle g \rangle = \{ \cdots g^{-2}, g^{-1}, 1, g, g^2, \cdots \}.$$

And by (e) any of two of such elements are distinct.

$\blacksquare$

<u>Corollary 3.26</u>. The order of an element g coincides with the cardinal of the cyclic subgroup $\langle g \rangle$ generated by g.

<u>Example</u>. $\mathbb{Z}_8^* = \{ \bar{1}, \bar{3}, \bar{5}, \bar{7} \}$

All the elements $\neq \bar{1}$ have order 2.